

Parallel Expectation-Maximization for Clustering

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January 2026

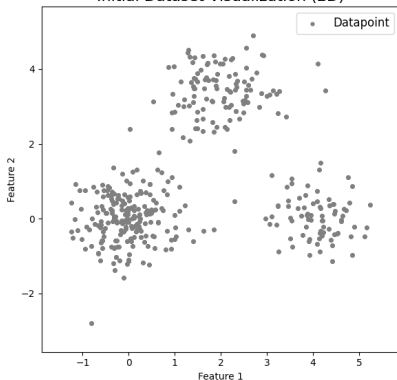


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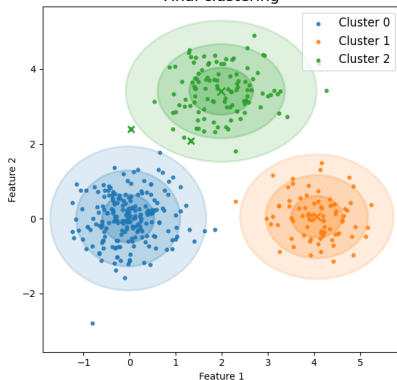
What is clustering - A Visual Example

Initial Dataset Visualization (2D)



First iteration of the clustering process

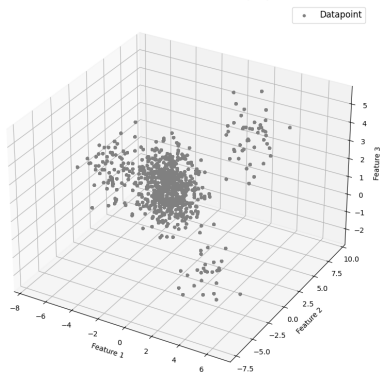
Final clustering



Final clustering result

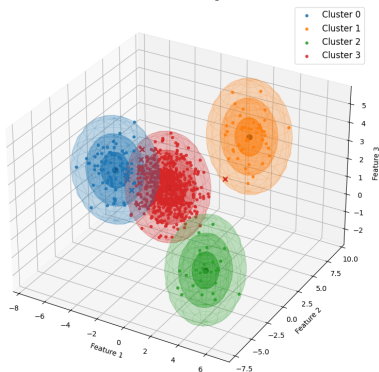
What is clustering - A Visual Example (3D)

Initial Dataset Visualization (3D)



First iteration (3D view)

Final Clustering



Final clustering (3D view)

What is clustering - More Formally

- Partition a dataset $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ where $\mathbf{x}_i \in \mathbb{R}^D$ into K groups (clusters).
- Each cluster is represented by a probability distribution with parameters θ_k .
- Find the cluster parameters that maximize the log-likelihood of the data:

$$\max_{\theta} (\mathcal{L}(\theta | \mathcal{X}))$$

$$\mathcal{L}(\theta | \mathcal{X}) = \log \sum_{i=1}^N \left(\sum_{k=1}^K \pi_k p(\mathbf{x}_i | \theta_k) \right)$$

Gaussian Mixture Models

- Each cluster is represented by a Gaussian distribution with its own mean and covariance:

$$p(\mathbf{x} \mid \theta) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k)$$

$$\sum_{k=1}^K \pi_k = 1, \quad \pi_k \geq 0$$

- The likelihood of a data point $\mathbf{x}$ to belong to cluster k is given by:

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k) = \frac{1}{(2\pi)^{D/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right)$$

Gaussian Mixture Models – Approximation

- We assumed the features are uncorrelated within each component, simplifying the covariance matrix to a diagonal form:

$$\Sigma_k = \begin{pmatrix} \sigma_{k,1} & 0 & \cdots & 0 \\ 0 & \sigma_{k,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{k,D} \end{pmatrix}$$

- This implies that the likelihood can be expressed as:

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k) = \frac{1}{(2\pi)^{D/2} \prod_{d=1}^D \sigma_{k,d}^{1/2}} \exp \left(-\frac{1}{2} \sum_{d=1}^D \frac{(x_d - \mu_{k,d})^2}{\sigma_{k,d}} \right)$$

Chicken and Egg Problem

- To cluster the data, we need to know the parameters of the Gaussian components.
- To estimate the parameters, we need to know the cluster assignments of the data points.
- This circular dependency is known as the "chicken and egg" problem in clustering.

The solution - EM-Algorithm

The Expectation-Maximization (EM) algorithm is an iterative method to find maximum likelihood estimates of parameters in models with latent variables.

EM-Algorithm - Steps

The EM algorithm consists of two main steps:

- **E-step (Expectation step):** Compute the expected value of the latent variables given the current parameter estimates.
- **M-step (Maximization step):** Update the parameters to maximize the expected log-likelihood found in the E-step.

EM-Algorithm - Pseudocode

Algorithm 1 Expectation-Maximization for GMM clustering

```
1: Randomly initialize parameters  $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$ 
2: for  $i = 1 \dots \text{MAX\_ITER}$  do
3:   E-step: compute responsibilities
4:   M-step: update parameters
5:   Compute log-likelihood  $\mathcal{L}$ 
6:   if Converged then
7:     break
8:   end if
9: end for
10: Clustering: assign labels
```

EM-Algorithm - E-Step

Using the current parameter estimates θ , we compute the responsibilities, i.e. the posterior probability that each data point belongs to each Gaussian component.

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j)}$$

EM-Algorithm - M-Step

Using the responsibilities γ_{ik} computed in the E-step, we update the parameters:

- **Sum of responsibilities for each component:** $N_k = \sum_{i=1}^N \gamma_{ik}$
- **Update of means:** $\mu_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} \mathbf{x}_i$
- **Update of covariances:** $\Sigma_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} (\mathbf{x}_i - \mu_k)(\mathbf{x}_i - \mu_k)^T$
- **Update of mixture weights:** $\pi_k = \frac{N_k}{N}$

Convergence

The log-likelihood of the data given the model parameters is computed as:

$$\mathcal{L}(\theta) = \sum_{i=1}^N \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k) \right)$$

Convergence it's reached if the change is below a predefined threshold ϵ .

Clustering Assignment

After convergence, each data point is assigned to the cluster with the highest responsibility:

$$\hat{z}_i = \arg \max_k p(z_i = k \mid \mathbf{x}_i, \theta) = \arg \max_k \gamma_{ik}$$

Computational Complexity Analysis

The computational cost of the EM algorithm is determined by the number of data points N , clusters K , and feature dimensions D .

It results in an overall per-iteration complexity:

$$O(NKD)$$

Parallel Design

Implementation

State-of-the-Art

Our Approach

Parallelization Strategy - E-Step

Parallelization Strategy - M-Step

Performance Considerations

Hybrid Parallelization

Scalability

Conclusion

Thank you!